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Internal Rating and Risk-Based Loan Pricing: A Comparative Analysis

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Credit Scoring Variable Selection



Selection of Discriminatory Financial Indicators

Four financial indicators were initially evaluated for the discriminant credit scoring framework.

Customer Retention Rate (CRR) was excluded from the final specification due to its limited discriminatory power between healthy and distressed firms.

The final model therefore focused on Return on Sales (ROS), Leverage, and Interest Coverage Ratio (ICR), which showed stronger differentiation across credit quality groups and higher explanatory relevance for default risk assessment.

	ROS(%)	Leverage	ICR	CRR
Healthy Firms	0,86	0,04	1,40	0,12
Distressed Firms	0,56	0,24	1,07	0,12
HF Std Dev	0,17	0,28	0,92	0,05
DF Std Dev	0,31	0,21	0,73	0,04
HF Min	0,62	-0,38	0,02	0,02
DF Min	0,20	-0,17	0,19	0,05
HF Max	1,18	0,50	2,75	0,19
DF Max	1,20	0,50	1,83	0,18

Discriminant Analysis



Discriminant Weight Vector Estimation

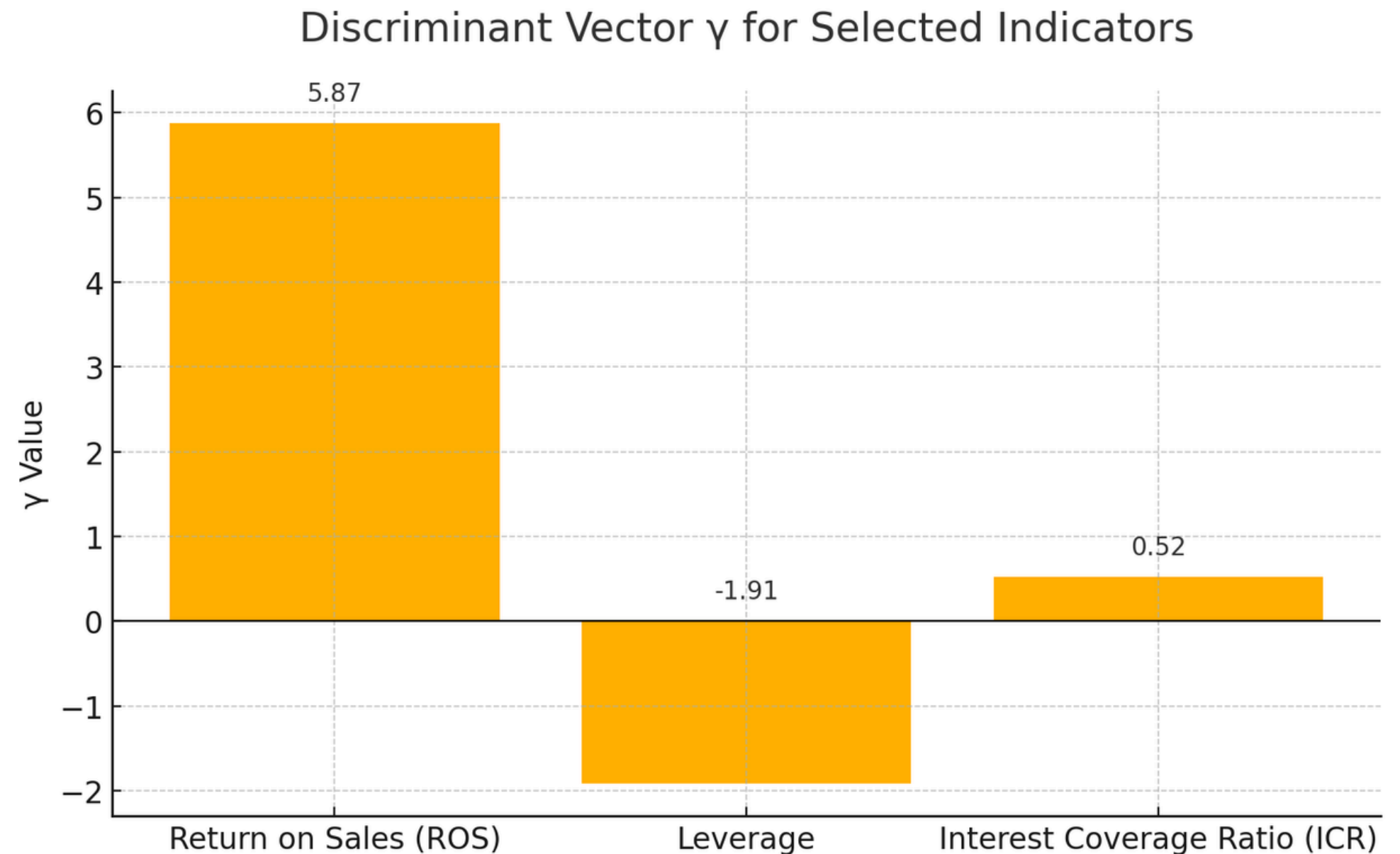
Linear Discriminant Analysis (LDA) was applied to the selected financial indicators: Return on Sales (ROS), Leverage, and Interest Coverage Ratio (ICR).

The discriminant weight vector γ was estimated using the inverse weighted covariance matrix and the difference between the mean vectors of healthy and distressed firms.

The resulting coefficients maximize the statistical separation between the two credit quality groups and define the contribution of each variable to the final creditworthiness score (Z-score).

Estimated γ vector coefficients:

- ROS: 5.87
- Leverage: -1.91
- ICR: 0.52



Model Performance: Type I & II Errors

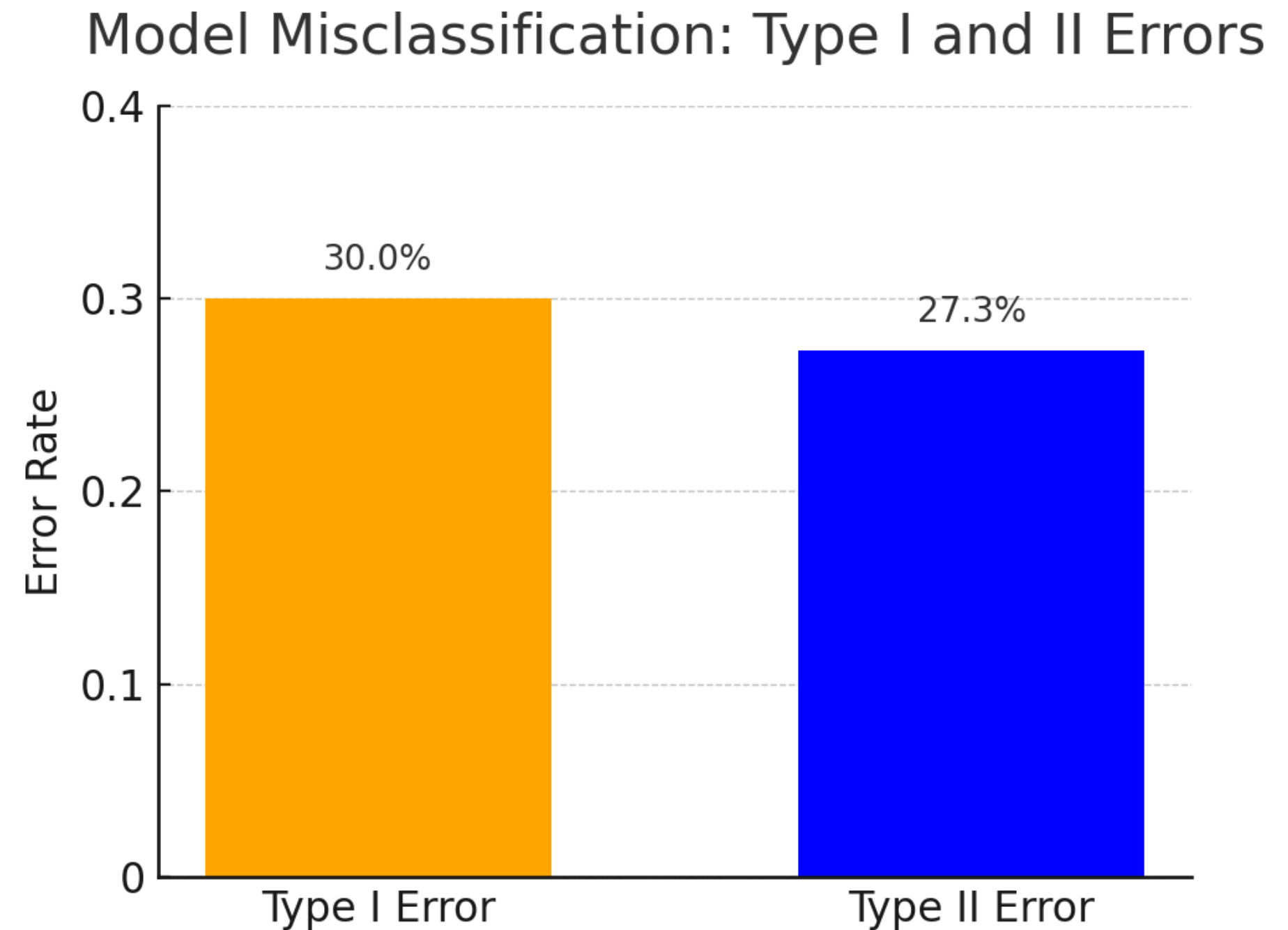


Classification Accuracy and Misclassification Analysis

A classification cutoff $\alpha = 4.538$ was defined as the midpoint between the average Z-scores of healthy and distressed firms. Using this threshold, the discriminant model produced the following in-sample misclassification rates:

- Type I Error: 3 out of 10 distressed firms were incorrectly classified as healthy (30.0%)
- Type II Error: 6 out of 22 healthy firms were incorrectly classified as distressed (27.3%)

The results highlight the trade-off between false negatives and false positives commonly observed in credit scoring models. From a risk management perspective, Type I errors are generally more critical, as they correspond to granting credit to financially distressed firms.



Internal Rating System and Loan Decision



A four-class internal rating framework was constructed using the empirical distribution of discriminant Z-scores.

Internal rating thresholds:

- A-rated: $z > 6.5$
- B-rated: $5 < z \leq 6.5$
- C-rated: $4 < z \leq 5$
- Defaulted: $z \leq 4$

The framework was then applied to the two loan applicants. ABC achieved a Z-score of 12.03 and was classified as A-rated, indicating strong financial quality and low estimated default risk.

XYZ achieved a Z-score of 4.02 and was classified as C-rated, reflecting weaker credit quality and higher financial risk.

Given the bank's capital constraints and relative risk profile of the applicants, the loan was granted exclusively to ABC.

Borrower	Z-score	Internal Rating	Loan Decision
ABC	12,03	A	Approved
XYZ	4,02	C	Not selected

Rating-Based Default Probability Analysis



Internal ratings were mapped to historical cumulative default rates published by S&P Global (2024).

Based on the internal A-rating assigned to ABC, the following cumulative probabilities of default were estimated:

- 1-Year PD = 0.05%
- 2-Year PD = 0.11%

Borrower	Rating	1-Year PD	2-Year PD
ABC	A	0,05%	0,11%

These results indicate a low expected default risk and support the strong credit quality assigned to ABC under the internal rating framework.

Global corporate average cumulative default rates, 1981-2024

	--Time horizon--														
(%)	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Y11	Y12	Y13	Y14	Y15
AAA	0.00	0.03	0.13	0.23	0.34	0.44	0.49	0.57	0.62	0.67	0.70	0.73	0.75	0.81	0.86
AA	0.02	0.05	0.11	0.19	0.28	0.37	0.45	0.52	0.59	0.65	0.71	0.76	0.81	0.86	0.90
A	0.05	0.11	0.19	0.29	0.39	0.51	0.65	0.78	0.90	1.03	1.14	1.25	1.35	1.45	1.56
BBB	0.14	0.38	0.67	1.01	1.36	1.71	2.00	2.30	2.58	2.86	3.13	3.35	3.56	3.78	4.01
BB	0.56	1.76	3.12	4.48	5.75	6.93	7.94	8.86	9.68	10.44	11.06	11.65	12.17	12.60	13.05
B	2.93	6.93	10.46	13.31	15.60	17.45	18.90	20.06	21.08	22.02	22.82	23.43	24.02	24.57	25.11
CCC/C	26.12	35.92	41.32	44.35	46.53	47.57	48.61	49.29	49.89	50.43	50.85	51.32	51.86	52.26	52.30
Investment grade	0.08	0.21	0.37	0.57	0.77	0.98	1.17	1.34	1.52	1.69	1.85	1.98	2.11	2.24	2.38
Speculative grade	3.54	6.78	9.55	11.79	13.64	15.15	16.39	17.41	18.32	19.15	19.85	20.44	20.99	21.47	21.93
All rated	1.50	2.91	4.13	5.15	6.00	6.72	7.31	7.81	8.26	8.67	9.02	9.31	9.58	9.83	10.07

Merton Model: Structural Default Estimation



The Merton (1973) structural credit risk model was applied to estimate ABC’s 2-year probability of default. Under the Merton framework, default occurs when the market value of the firm’s assets falls below the face value of debt at maturity.

The model estimates default risk using the distance-to-default measure d_2 and computes the risk-neutral probability of default as $N(-d_2)$.

The analysis incorporates the firm’s asset value, debt level, asset volatility, risk-free rate, and loan maturity.

Borrower	Equity	Debt	Asset Volatility	d1	d2	Risk-Neutral PD	Credit Decision
ABC	\$2,747M	\$3,700M	22%	2,235	1,924	2,72%	Approved

Market Comparables: Bond Spread Estimation



Starbucks Corporation was selected as a market comparable based on similarities in industry exposure, maturity profile, and investment-grade credit quality (BBB+).

Using the yield of the selected corporate bond and a 2-year U.S. Treasury rate of 4,00%, the implied credit spread was estimated at 66,2 basis points, resulting in an estimated loan yield of 4,662%.

Applied to a USD 100 million 2-year bullet loan, this corresponds to a total repayment amount of approximately USD 109,52 million at maturity.

Firm	Comparable Firm	Credit Rating	Spread	Risk-Free Rate	Implied Loan Yield	Repayment at Maturity
ABC	Starbucks	BBB+	0,662%	4,00%	4,662%	\$109,52M

Comparative Assessment of Credit Risk Methodologies



Rating-Based Approach

Simple and transparent, but based on historical averages

→ 2-Year PD: 0,11%

Merton Model

Forward-looking and theoretically robust, but highly sensitive to input assumptions

→ Risk-Neutral PD (2Y): 2,72%

Market-Based Approach

Reflects current market conditions, but may incorporate liquidity and market sentiment effects

→ Spread: 0,662% | Implied Yield: 4,662%

Final Credit Decision

Overall, the three approaches consistently support the approval of ABC's loan application, despite differences in model sensitivity and methodology.

Based on the combined evidence from internal ratings, structural credit risk analysis, and market comparables, ABC was assigned an A-rating and an implied loan yield of 4,662%.



Thank you!

For further information or discussion regarding the analysis,
please contact:

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